

Matrix - Complete Notes (Short & Clear)

1. Square Matrix

Definition: Row = Column ($n \times n$)

Example:

[1 2]

[3 4]

2. Rectangular Matrix

Definition: Row $\neq$ Column

Example (2x3):

[1 2 3]

[4 5 6]

3. Row Matrix

Definition: Only 1 row ($1 \times n$)

Example:

[5 7 9]

4. Column Matrix

Definition: Only 1 column ($n \times 1$)

Example:

[2]

[4]

[6]

5. Zero Matrix

Definition: All elements = 0

Example:

$$\begin{bmatrix} 0 & 0 \end{bmatrix}$$
$$\begin{bmatrix} 0 & 0 \end{bmatrix}$$

6. Diagonal Matrix

Main diagonal ■■■■ ■■ 0

Example:

$$\begin{bmatrix} 1 & 0 & 0 \end{bmatrix}$$
$$\begin{bmatrix} 0 & 4 & 0 \end{bmatrix}$$
$$\begin{bmatrix} 0 & 0 & 7 \end{bmatrix}$$

7. Identity Matrix (I)

Diagonal elements = 1

Example:

$$\begin{bmatrix} 1 & 0 \end{bmatrix}$$
$$\begin{bmatrix} 0 & 1 \end{bmatrix}$$

8. Scalar Matrix

Diagonal elements equal

Example:

$$\begin{bmatrix} 5 & 0 \end{bmatrix}$$
$$\begin{bmatrix} 0 & 5 \end{bmatrix}$$

9. Upper Triangular Matrix

Below diagonal = 0

Example:

[1 2 3]

[0 4 5]

[0 0 6]

10. Lower Triangular Matrix

Above diagonal = 0

Example:

[1 0 0]

[2 3 0]

[4 5 6]

Transpose (A^T)

Row and Column interchange

Example:

$A = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}$

$A^T = \begin{bmatrix} 1 & 3 \\ 2 & 4 \end{bmatrix}$

Determinant (2x2)

$|A| = ad - bc$

Inverse (2x2)

$$A^{-1} = \frac{1}{(ad-bc)} * \begin{bmatrix} d & -b \\ -c & a \end{bmatrix}$$

Power of Matrix

$$A^2 = A \times A \text{ (Only for square matrix)}$$

Short Summary Table

Type	Condition
Square	Row = Column
Rectangular	Row != Column
Row Matrix	1 Row
Column Matrix	1 Column
Zero	All 0
Diagonal	Non-zero only in diagonal
Identity	Diagonal = 1
Scalar	Diagonal equal
Upper Triangular	Below diagonal = 0
Lower Triangular	Above diagonal = 0